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Multiaxial Stress Interaction and Failure Initiation in Internally Pressurized Thick-Walled Carbon/Epoxy Composite Cylinders

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Abstract: The strength assessment of thick-walled composite cylinders under internal pressure requires consideration of the nonuniform stress distribution and the interaction of stress components arising from material orthotropy and axial restraint. Conventional assessments based primarily on circumferential stress may not adequately represent the multiaxial stress state governing failure initiation. This study investigates the elastic stress response and initial failure of an internally pressurized thick-walled cylinder made of Hexcel AS4/3501-6 carbon fiber/epoxy composite, with particular attention to the effects of wall thickness and three-dimensional stress interaction. An analytical formulation was developed using orthotropic elasticity under plane-strain conditions, and the radial, circumferential, and axial normal stresses were determined across the cylinder wall. The three-dimensional Tsai–Wu criterion was then applied to evaluate the combined influence of these stresses on failure initiation and to determine the allowable internal pressure. The calculations showed a nonuniform stress distribution, with the circumferential stress reaching its maximum at the inner surface and decreasing toward the outer wall. For the representative configuration, the maximum circumferential stress was approximately 188 MPa, while the axial stress remained nonzero because of the imposed axial constraint. Variations in wall thickness produced substantial changes in the stress distribution and the predicted failure response. The analysis also established a procedure for identifying the pressure at which the Tsai–Wu failure threshold is first reached. These findings clarify the role of orthotropic elasticity and multiaxial stress interaction in the strength assessment of pressurized composite cylinders. The analytical framework provides a basis for evaluating initial failure margins and examining the influence of wall thickness on structural pressure capacity.

Keywords: Thick-walled composite cylinder; Orthotropic elasticity; Multiaxial stress interaction; Plane-strain analysis; Tsai–Wu failure criterion; Failure initiation

1 Introduction

The structural integrity of cylindrical pressure vessels and piping systems is a major consideration in aerospace, oil and gas, chemical processing, and defense engineering, where pressurized components must sustain substantial mechanical loads while meeting stringent requirements for weight, reliability, and operational safety. Carbon-fiber-reinforced polymer (CFRP) composites are widely considered for such applications because of their high specific stiffness, high strength-to-weight ratio, and considerable flexibility in structural design. Their mechanical response, however, differs fundamentally from that of conventional isotropic materials. The directional dependence of elastic stiffness and strength produces nonuniform stress distributions under internal pressure, and failure may be governed by the combined action of several stress components rather than by the maximum value of an individual component.

Accurate characterization of the stress field and the conditions governing failure initiation is therefore essential for assessing the pressure resistance of composite cylindrical structures [1–3].

The mechanical analysis of fiber-reinforced composites has been developed through several complementary theoretical approaches, including anisotropic elasticity, micromechanics, constitutive modeling, strength criteria, and laminated structural theories. Classical laminate theory and first-order and higher-order shear deformation theories provide established descriptions of the deformation and stress response of composite plates and shells, particularly when the effects of transverse shear deformation and laminate configuration are important. Micromechanical formulations, meanwhile, relate the effective material properties to the characteristics of the reinforcing fibers and surrounding matrix. Probabilistic approaches have also been introduced to account for uncertainties associated with constituent properties and the resulting variability in composite stiffness. These developments provide the theoretical basis for examining the mechanical behavior of composite structures, although their applicability depends on the structural geometry, material representation, and loading conditions under consideration [4–7].

The response of composite cylindrical structures to internal and external pressure has received considerable attention in analytical, numerical, and experimental investigations. Previous studies have examined the effects of ply orientation, stacking sequence, fiber angle, elastic anisotropy, environmental conditions, and geometric parameters on stress distributions and failure initiation. The results have established that the reinforcement architecture can substantially alter the distribution of stresses and the location at which the first failure condition is reached [8–10]. Further investigations of filament-wound and thick-walled composite cylinders have addressed localized damage, through-thickness stress variations, and the sensitivity of structural strength to the arrangement and orientation of reinforcing fibers [11, 12]. Three-dimensional finite-element formulations and established composite failure criteria have also been employed to assess the strength of pressure vessels, including composite structures developed for high-pressure hydrogen storage. Such analyses have demonstrated the importance of accounting for the multiaxial stress state when evaluating structural performance under pressure [13, 14].

In addition to elastic stress analysis, the mechanical response of composite pressure-bearing structures may be influenced by loading history, manufacturing conditions, and material heterogeneity. Previous investigations have considered strain-rate dependence, viscoelastic deformation, residual stresses introduced during curing, variations in fiber volume fraction, and manufacturing defects such as voids and imperfect interfaces [15–17]. Thermoviscoelastic effects, orientation-dependent strength characteristics, and damage associated with impact or compressive loading have likewise been examined in the broader literature on composite materials [18–21]. These studies indicate that the structural response of fiber-reinforced composites may involve mechanisms extending beyond instantaneous elastic deformation. Nevertheless, an elastic formulation remains useful for isolating the influence of material orthotropy, geometric configuration, and mechanical constraints on the stress state preceding initial failure. The present investigation adopts this restricted mechanical setting; environmental degradation, residual stresses, time-dependent material behavior, and progressive damage are not included in the governing model.

For a thick-walled orthotropic composite cylinder, the interaction between material directionality and structural constraints requires particular attention. Unlike a thin-walled approximation, which may neglect radial stress under suitable geometric and loading conditions, a thick-walled formulation must account for the variation of stress components across the cylinder wall. Furthermore, when axial deformation is constrained, the plane-strain condition introduces a nonzero axial stress even though the corresponding axial strain vanishes. The resulting stress state contains radial, circumferential, and axial normal components whose relative magnitudes depend on the orthotropic elastic properties and cylinder geometry. Assessing structural strength solely from the circumferential stress may therefore be insufficient, particularly when the material exhibits different tensile and compressive strengths along its principal directions. Although analytical elasticity solutions and composite failure criteria are well established, their combined application requires careful treatment of the constitutive assumptions, three-dimensional stress interaction, and the distinction between initial failure and subsequent structural collapse. In this context, the present study combines an analytical plane-strain formulation with the three-dimensional Tsai–Wu failure criterion to examine how the through-thickness stress distribution contributes to failure initiation under internal pressure. Particular attention is given to the influence of wall thickness on the predicted failure index and the pressure at which the initial failure threshold is reached. This approach provides a consistent basis for examining the relationship between orthotropic elastic response, multiaxial strength assessment, and the pressure capacity of composite cylindrical components.

Accordingly, the objective of this study is to investigate multiaxial stress interaction and failure initiation in a thick-walled cylinder manufactured from Hexcel AS4/3501-6 carbon fiber/epoxy composite and subjected to uniform internal pressure. The composite is represented as a homogeneous orthotropic continuum, and analytical expressions for radial displacement and the radial, circumferential, and axial normal stresses are obtained under plane-strain conditions. These stress components are subsequently incorporated into the three-dimensional Tsai–Wu criterion to evaluate the variation of the failure index across the cylinder wall and examine the influence of wall thickness on the predicted allowable pressure. The analysis focuses on the relationship between the nonuniform elastic stress

field and the onset of material failure, rather than on the evolution of damage after the initial failure condition has been reached. The resulting formulation is intended to support the assessment of initial failure margins and the interpretation of wall-thickness effects in composite pressure-bearing structures. Its conclusions are limited to the assumed elastic material response, uniform internal loading, and constrained axial deformation; progressive failure, ultimate burst pressure, and long-term environmental durability remain outside the scope of the investigation.

2 Problem Formulation

Figure 1 illustrates the cross-sectional geometry of a long, thick-walled cylindrical tube made of Hexcel AS4/3501-6 carbon fiber/epoxy composite. The inner and outer radii are denoted by a and b, respectively, and the wall thickness is defined as $h = b - a$. A uniform internal pressure p is applied to the inner cylindrical surface, while the outer surface is traction-free. For generality, both internal and external pressures are included in the analytical derivation. All numerical calculations presented in this study are restricted to internal pressure loading, with the external pressure set to zero.

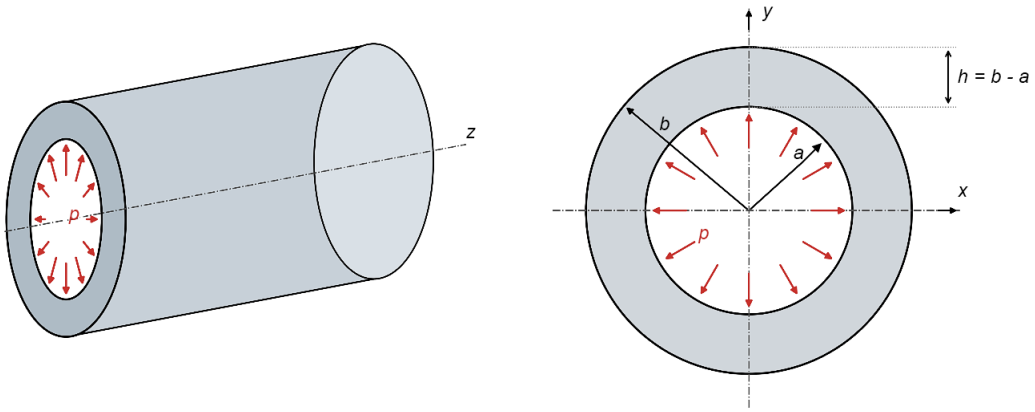


Figure 1. Schematic representation of the thick-walled composite cylinder and the applied pressure boundary conditions

The cylinder is represented as a homogeneous, linearly elastic orthotropic continuum whose principal material directions coincide with the radial (r), circumferential (θ), and axial (z) coordinates. The effective elastic properties are assumed to remain constant through the wall thickness. Under axisymmetric loading, the radial displacement depends only on the radial coordinate, $u = u(r)$, and the circumferential displacement vanishes. The long-cylinder approximation, together with the assumed axial restraint, permits the adoption of a plane-strain formulation in which the axial strain is zero. Nevertheless, the axial stress generally remains nonzero because of the constitutive coupling among the normal stress components. Its contribution is therefore retained in the subsequent three-dimensional strength assessment.

Under the plane-strain condition, $\varepsilon_z = 0$, the radial and circumferential strains are expressed as $\varepsilon_r = du/dr$ and $\varepsilon_\theta = u/r$, respectively. The corresponding constitutive relations for the orthotropic cylinder are written as

$$\sigma_r = C_{11}\frac{du}{dr} + C_{12}\frac{u}{r} \tag{1}$$

$$\sigma_\theta = C_{12}\frac{du}{dr} + C_{22}\frac{u}{r} \tag{2}$$

$$\sigma_z = v_{zr}\sigma_r + v_{zv}\sigma_\theta \tag{3}$$

The Poisson's ratio v_{ij} is defined as the negative ratio of the transverse strain in direction j to the axial strain produced by uniaxial loading in direction i. This convention is used consistently throughout the constitutive formulation and numerical calculations.

$$C_{11} = \frac{\frac{1}{E_\theta} - \frac{v_{z\theta}^2}{E_z}}{\left(\frac{1}{E_r} - \frac{v_{zr}^2}{E_z}\right)\left(\frac{1}{E_\theta} - \frac{v_{z\theta}^2}{E_z}\right) - \left(\frac{v_{r\theta}}{E_r} + \frac{v_{zr}v_{z\theta}}{E_z}\right)^2},$$

$$C_{22} = \frac{\frac{1}{E_r} - \frac{v_{zr}^2}{E_z}}{\left(\frac{1}{E_r} - \frac{v_{zr}^2}{E_z}\right)\left(\frac{1}{E_\theta} - \frac{v_{z\theta}^2}{E_z}\right) - \left(\frac{v_{r\theta}}{E_r} + \frac{v_{zr}v_{z\theta}}{E_z}\right)^2},$$

$$C_{12} = \frac{\frac{v_{r\theta}}{E_\theta} + \frac{v_{zr}v_{z\theta}}{E_z}}{\left(\frac{1}{E_r} - \frac{v_{zr}^2}{E_z}\right)\left(\frac{1}{E_\theta} - \frac{v_{z\theta}^2}{E_z}\right) - \left(\frac{v_{r\theta}}{E_r} + \frac{v_{zr}v_{z\theta}}{E_z}\right)^2}.$$

where, E_r, E_θ, and E_z denote the Young's moduli along the principal material directions. The symmetry of the elastic compliance matrix requires the reciprocal relationships

$$\frac{v_{r\theta}}{E_r} = \frac{v_{\theta r}}{E_\theta}; \quad \frac{v_{rz}}{E_r} = \frac{v_{zr}}{E_z}; \quad \frac{v_{\theta z}}{E_\theta} = \frac{v_{z\theta}}{E_z}.$$

These conditions ensure consistency between the directional elastic properties and the underlying strain-energy formulation. The elastic constants must additionally satisfy the positive-definiteness requirements associated with a stable orthotropic material.

CFRP cylinders commonly employ reinforcement arrangements containing circumferential and axial fiber orientations to resist the principal loads generated by internal pressure. Circumferential reinforcement primarily contributes to hoop stiffness and strength, whereas axial reinforcement supports longitudinal loading. The radial response is influenced more strongly by the matrix and interlaminar properties because reinforcement in the through-thickness direction is generally limited. Consequently, the effective elastic moduli, shear moduli, and Poisson's ratios may differ among the three principal directions, giving rise to orthotropic mechanical behavior [2].

In the present formulation, the individual composite plies are not modeled separately. Instead, the cylinder is treated as an equivalent homogeneous orthotropic continuum, with the collective influence of the reinforcement architecture represented by effective material constants. This representation permits analytical examination of the through-thickness stress field without explicitly resolving individual ply interfaces. It is therefore appropriate for the present continuum-level analysis, provided that the effective elastic properties adequately describe the composite configuration under consideration.

For axisymmetric deformation in the absence of body forces, radial equilibrium requires

$$\frac{d\sigma_r}{dr} = \frac{\sigma_\theta - \sigma_r}{r} \tag{4}$$

Substitution of Eqs. (1) and (2) into Eq. (4), together with the assumption of radially uniform elastic properties, yields the following Euler-type differential equation:

$$C_{11}\frac{d^2u}{dr^2} + \frac{C_{11}}{r}\frac{du}{dr} - \frac{C_{22}}{r^2}u = 0 \tag{5}$$

The characteristic equation admits two roots of opposite signs. Accordingly, the general solution for the radial displacement is:

$$u(r) = Ar^m + Br^{-m} \tag{6}$$

where, $m = \sqrt{C_{22}/C_{11}}$.

The parameter m characterizes the influence of orthotropic stiffness on the radial variation of displacement and stress. For a mechanically admissible material with positive effective stiffnesses, m is real and positive. The constants A and B are determined from the prescribed radial traction conditions.

Substituting Eq. (6) into the constitutive relations gives:

$$\sigma_r = (C_{11}m + C_{12})\,Ar^{m-1} + (C_{12} - C_{11}m)\,Br^{-m-1} \tag{7}$$

$$\sigma_\theta = (C_{12}m + C_{22})\,Ar^{m-1} + (C_{22} - C_{12}m)\,Br^{-m-1} \tag{8}$$

These expressions show that the radial and circumferential stresses depend on the effective orthotropic stiffness coefficients and the radial position. Their distributions are therefore governed jointly by the material properties, geometric dimensions, and imposed pressure boundary conditions.

For the general case involving internal pressure p and external pressure q, the radial stress must satisfy

$$\sigma_r(a) = -p; \sigma_r(b) = -q \tag{9}$$

The negative signs follow the convention that tensile normal stresses are positive, whereas the applied pressures produce compressive radial tractions. In the present investigation, the external surface is traction-free, so $q = 0$.

For compactness, the following stiffness ratio is introduced: $\lambda = \frac{C_{12}}{C_{11}}$.

Applying Eq. (9) to the stress distribution in Eq. (7) gives:

$$A = \frac{pa^{m+1} - qb^{m+1}}{C_{11}\left(b^{2m} - a^{2m}\right)(m+\lambda)}; B = \frac{pa^{2m}b^{m+1} - qa^{m+1}b^{2m}}{C_{11}\left(b^{2m} - a^{2m}\right)(m-\lambda)}$$

The expressions for the integration constants are valid provided that $m+\lambda \neq 0$ and $m-\lambda \neq 0$. These conditions are satisfied for the material parameters considered in the present analysis. Substitution into Eq. (6) provides the radial displacement:

$$u(r) = \frac{1}{\left(b^{2m} - a^{2m}\right)C_{11}}\left[\frac{pa^{m+1} - qb^{m+1}}{m+\lambda}r^{m} + \frac{pa^{2m}b^{m+1} - qa^{m+1}b^{2m}}{m-\lambda}r^{-m}\right] \tag{10}$$

Using the displacement solution in Eqs. (7) and (8), the radial stress is obtained as:

$$\sigma_r(r) = \frac{pa^{m+1} - qb^{m+1}}{b^{2m} - a^{2m}}r^{m-1} - \frac{pa^{m+1}b^{2m} - qa^{2m}b^{m+1}}{b^{2m} - a^{2m}}r^{-m-1} \tag{11}$$

while the circumferential stress becomes:

$$\sigma_\theta(r) = m\left[\frac{pa^{m+1} - qb^{m+1}}{b^{2m} - a^{2m}}r^{m-1} + \frac{pa^{m+1}b^{2m} - qa^{2m}b^{m+1}}{b^{2m} - a^{2m}}r^{-m-1}\right] \tag{12}$$

The axial stress is subsequently determined from the plane-strain relation in Eq. (3):

$$\sigma_z(r) = \frac{\Gamma_1\left(pa^{m+1} - qb^{m+1}\right)}{b^{2m} - a^{2m}}r^{m-1} - \frac{\Gamma_2\left(pa^{m+1}b^{2m} - qa^{2m}b^{m+1}\right)}{b^{2m} - a^{2m}}r^{-m-1} \tag{13}$$

where, $\Gamma_1 = v_{zr} + mv_{z\theta}; \Gamma_2 = v_{zr} - mv_{z\theta}$.

Eq. (13) makes explicit the contribution of the radial and circumferential stress components to the axial stress under constrained axial deformation. Although the axial strain vanishes, the resulting axial stress must be included when evaluating the three-dimensional failure response of the orthotropic composite.

To facilitate the comparison of cylinders with different wall thicknesses, the governing expressions are rewritten using the following dimensionless geometric parameters:

$$\lambda = \frac{C_{12}}{C_{11}}; \beta_0 = \frac{a}{b}; \rho = \frac{r}{b}, (\beta_0 \leq \rho \leq 1) \tag{14}$$

where, β_0 is the inner-to-outer radius ratio and ρ is the normalized radial coordinate. An increase in β_0, with the outer radius held constant, corresponds to a reduction in wall thickness.

In terms of these parameters, the radial displacement is expressed as:

$$u(\rho) = \frac{b}{\left(1 - \beta_0^{2m}\right)C_{11}}\left[\frac{p\beta_0^{m+1} - q}{m+\lambda}\rho^{m} + \frac{p\beta_0^{m+1} - q\beta_0^{2m}}{m-\lambda}\rho^{-m}\right] \tag{15}$$

$$\sigma_r(\rho) = \frac{1}{1 - \beta_0^{2m}}\left[\left(p\beta_0^{m+1} - q\right)\rho^{m-1} - \left(p\beta_0^{m+1} - q\beta_0^{2m}\right)\rho^{-m-1}\right] \tag{16}$$

$$\sigma_\theta(\rho) = \frac{m}{1 - \beta_0^{2m}}\left[\left(p\beta_0^{m+1} - q\right)\rho^{m-1} + \left(p\beta_0^{m+1} - q\beta_0^{2m}\right)\rho^{-m-1}\right] \tag{17}$$

$$\sigma_z(\rho) = \frac{1}{1 - \beta_0^{2m}}\left[\Gamma_1\left(p\beta_0^{m+1} - q\right)\rho^{m-1} - \Gamma_2\left(p\beta_0^{m+1} - q\beta_0^{2m}\right)\rho^{-m-1}\right] \tag{18}$$

When the effective radial and circumferential stiffness coefficients are equal, the characteristic exponent reduces to unity, and the radial and circumferential stress distributions recover the classical Lamé solution. This limiting case provides a consistency check for the analytical stress formulation. Recovery of the complete isotropic constitutive response additionally requires the corresponding isotropic relations among the elastic constants.

3 Theoretical Model

Under the plane-strain conditions adopted in the present analysis, the axial strain is constrained to zero, whereas the axial stress generally remains nonzero because of the constitutive interaction between the radial, circumferential, and axial directions. Consequently, the stress state of the orthotropic thick-walled cylinder cannot be adequately characterized by the radial and circumferential components alone. Although the applied loading is axisymmetric, the axial constraint introduces an additional normal stress that must be considered when evaluating the onset of material failure. With the material principal axes aligned with the cylindrical coordinates, the plane-strain condition and the vanishing shear stresses are expressed as $\varepsilon_z = 0, \sigma_z \neq 0, \tau_{r\theta} = \tau_{\theta z} = \tau_{zr} = 0$.

The resulting three-dimensional normal stress state is evaluated using the Tsai–Wu failure criterion.

Failure assessment in fiber-reinforced composites requires consideration of the directional dependence of material strength and the interaction of simultaneously acting stress components. The maximum stress and maximum strain criteria evaluate individual stress or strain components against their corresponding allowable values. Although these criteria are useful for identifying component-wise strength limits, they do not explicitly account for interactions among different stress components. The Tsai–Wu criterion provides an alternative phenomenological formulation in which the combined stress state is represented by a quadratic function. Its linear terms distinguish between tensile and compressive strength limits, while its quadratic and cross-interaction terms account for the influence of multiaxial loading on the predicted failure threshold. The criterion is therefore suitable for examining initial failure in the orthotropic composite cylinder considered here, provided that the required strength parameters and interaction coefficients are appropriately specified.

For a general anisotropic material, the Tsai–Wu failure criterion can be expressed in tensor notation as:

$$F_i\sigma_i + F_{ij}\sigma_i\sigma_j = 1(i, j = 1, 2, 3, \ldots\ldots, 6) \tag{19}$$

where, σ_i denotes the stress components expressed in the material principal coordinate system using Voigt notation, F_i represents the linear strength coefficients, and F_{ij} denotes the quadratic strength coefficients. Repeated indices imply summation. The linear coefficients account for differences between tensile and compressive strength limits, whereas the quadratic coefficients describe the contributions of individual stress components and their interactions. The equality in Eq. (19) defines the predicted initial failure surface in stress space.

For the axisymmetric stress state considered in this study, the normal stresses are identified as $\sigma_1 = \sigma_\theta, \sigma_2 = \sigma_r, \sigma_3 = \sigma_z$.

Since the shear stresses vanish, the three-dimensional Tsai-Wu failure index is written as:

$$FI = F_1\sigma_\theta + F_2\sigma_r + F_3\sigma_z + F_{11}\sigma_\theta^2 + F_{22}\sigma_r^2 + F_{33}\sigma_z^2 + 2F_{12}\sigma_\theta\sigma_r + 2F_{23}\sigma_r\sigma_z + 2F_{13}\sigma_\theta\sigma_z \tag{20}$$

where, FI denotes the failure index, which characterizes the calculated stress state relative to the failure threshold defined by the adopted strength criterion. Three conditions are distinguished: $FI < 1$.

The calculated stress state remains below the Tsai-Wu initial failure threshold: $FI = 1$.

The calculated stress state reaches the predicted threshold for initial material failure: $FI > 1$.

The calculated stress state exceeds the prescribed failure threshold, indicating that the loading condition is no longer admissible according to the adopted criterion.

These conditions describe the prediction of initial failure and should not be interpreted as a direct representation of progressive damage, complete loss of structural load-carrying capacity, or ultimate burst pressure. In particular, the Tsai–Wu criterion does not independently describe the evolution of damage after the initial failure threshold has been reached.

The strength coefficients are defined using the tensile and compressive strengths along the three orthotropic material directions. In the present coordinate convention, X_T and X_C denote the tensile and compressive strengths in the circumferential direction, Y_T and Y_C are the corresponding strengths in the radial direction, and Z_T and Z_C represent the axial strength limits. All strength values are taken as positive magnitudes.

The linear and diagonal quadratic coefficients are expressed as:

$$\begin{array}{ll} F_1 = \frac{1}{X_T} - \frac{1}{X_C}, & F_{11} = \frac{1}{X_T X_C}, \\ F_2 = \frac{1}{Y_T} - \frac{1}{Y_C}, & F_{22} = \frac{1}{Y_T Y_C}, \\ F_3 = \frac{1}{Z_T} - \frac{1}{Z_C}, & F_{33} = \frac{1}{Z_T Z_C}. \end{array}$$

The interaction coefficients are required to account for the combined effects of normal stresses acting along different material directions. In the absence of independently measured multiaxial interaction parameters, the following estimates are adopted: $F_{12} = -0.5\sqrt{F_{11}F_{22}}, F_{23} = -0.5\sqrt{F_{22}F_{33}}, F_{13} = -0.5\sqrt{F_{11}F_{33}}$.

These expressions provide a prescribed approximation for the interaction coefficients rather than a direct experimental characterization of the multiaxial failure surface of Hexcel AS4/3501-6. Their use allows the influence

of the three normal stress components to be examined within a consistent analytical framework. Nevertheless, the predicted failure index and allowable pressure remain dependent on the selected interaction coefficients, and this dependence should be considered when interpreting the results.

The distinction between tensile and compressive strength is particularly important for carbon/epoxy composites because the material response is generally asymmetric under loading of opposite signs. The differences between the corresponding strength limits are represented by $X_C \neq X_T, Y_C \neq Y_T, Z_C \neq Z_T$.

The linear terms in Eq. (20), $F_1\sigma_\theta, F_2\sigma_r, F_3\sigma_z$ account for the tensile-compressive asymmetry of the material. The quadratic terms describe the contributions of the individual normal stresses, while the cross-interaction terms modify the predicted failure index according to the simultaneous action and relative signs of the stress components. Consequently, two stress states with similar circumferential stresses may produce different failure indices when their radial or axial stresses differ.

The Tsai–Wu criterion also differs from failure models that distinguish individual damage mechanisms. Whereas criteria such as Hashin and Puck are formulated to identify particular fiber- or matrix-related failure modes, the Tsai–Wu expression provides a single scalar measure of the combined stress state. Its continuous quadratic form is convenient for analytical calculations and numerical implementation. However, the resulting failure index does not identify the physical damage mechanism responsible for failure initiation. For the present study, the criterion is used to determine the onset-of-failure threshold rather than to classify individual damage modes or predict subsequent damage development.

The axial stress contribution is particularly relevant to thick-walled composite cylinders subjected to constrained axial deformation. Under plane-strain conditions, the nonzero axial stress is determined by the orthotropic constitutive relations and depends on the radial and circumferential stresses. Its contribution to Eq. (20) is represented by the linear term $F_3\sigma_z$, the quadratic term $F_{33}\sigma_z^2$, and the interaction terms $2F_{23}\sigma_r\sigma_z$ and $2F_{13}\sigma_\theta\sigma_z$.

These terms allow the effect of axial restraint on the predicted initial failure condition to be evaluated without reducing the analysis to a circumferential-stress criterion.

It should be emphasized that the interaction terms describe the mathematical dependence of the selected failure criterion on the combined stress state. They do not, by themselves, establish a hydrostatic-pressure strengthening mechanism or quantify matrix plasticity and confinement-induced damage resistance. Such mechanisms would require additional constitutive assumptions or experimental evidence beyond the elastic model adopted here.

By combining the analytical stress distributions obtained in Section 2 with the three-dimensional Tsai-Wu criterion, the failure index can be evaluated as a function of radial position and internal pressure. For a specified cylinder geometry and set of orthotropic material properties, the pressure corresponding to initial failure is determined by the first attainment of the condition $\max_{a\leq r\leq b} FI\,(r, p_{\mathrm{cr}}) = 1$ as the internal pressure increases from zero.

Here, p_{cr} denotes the predicted critical internal pressure associated with initial failure. This procedure permits the influence of wall thickness and multiaxial stress interaction on the predicted failure threshold to be examined within the assumptions of linear orthotropic elasticity. The resulting pressure represents an analytical estimate of initial failure capacity and should be distinguished from the ultimate pressure resistance of a progressively damaged composite cylinder.

4 Numerical Results and Discussion

The analytical formulation developed in Sections 2 and 3 was applied to a thick-walled cylindrical component made of Hexcel AS4/3501-6 carbon fiber/epoxy composite. This material system has been widely examined in composite mechanics and structural strength studies and provides a representative basis for investigating the influence of orthotropic elasticity on the stress response of pressure-bearing components.

The cylinder was represented by an equivalent homogeneous orthotropic material model. The elastic constants adopted for the numerical analysis are listed below. These parameters were treated as effective properties in the radial, circumferential, and axial directions of the cylinder [3–6]:

$$
\begin{aligned}
E_r &= 10.3\,\text{GPa},\\
E_\theta &= 142\,\text{GPa},\\
E_z &= 125\,\text{GPa},\\
G_{r\theta} &= 7.2\,\text{GPa},\\
G_{rz} &= 5.7\,\text{GPa},\\
G_{\theta z} &= 50\,\text{GPa}.
\end{aligned}
$$

The Poisson's ratios were specified as:

$$
\begin{aligned}
v_{\theta r} &= 0.27, \quad & v_{r\theta} &= 0.0196,\\
v_{zr} &= 0.29, \quad & v_{rz} &= 0.0239,\\
v_{z\theta} &= 0.035, \quad & v_{\theta z} &= 0.0398.
\end{aligned}
$$

The orthotropic compliance matrix was constructed using the reported elastic constants, with reciprocal symmetry imposed consistently with the adopted Poisson's ratio convention. Minor differences arising from the tabulated precision were accounted for in the numerical implementation.

The strength parameters required for the three-dimensional Tsai–Wu criterion were based on the reported tensile, compressive, and shear properties of Hexcel AS4/3501-6 [7, 22]. The adopted values were:

$$\begin{aligned} X_T &= 1950\,\text{MPa}, \\ X_C &= 1480\,\text{MPa}, \\ Y_T &= 49\,\text{MPa}, \\ Y_C &= 200\,\text{MPa}, \\ S &= 79\,\text{MPa}. \end{aligned}$$

where, X_T and X_C denote the tensile and compressive strength limits assigned to the circumferential material direction, while Y_T and Y_C represent the corresponding transverse strength limits assigned to the radial direction. The parameter S denotes the in-plane shear strength. Although shear strength does not enter the reduced normal-stress form of Eq. (20), it is included for completeness in the material description.

For the equivalent homogeneous orthotropic material model adopted in this study, the axial tensile and compressive strength limits were assumed to be equal to their corresponding circumferential values. Accordingly, the following strength parameters were adopted for the three-dimensional Tsai-Wu failure assessment: $Z_T = X_T = 1950$ MPa; $Z_C = X_C = 1480$ MPa.

The numerical analysis considered a long, thick-walled cylinder subjected to uniform internal pressure, with no external pressure applied. Accordingly, the loading and geometric conditions were expressed as $q = 0, \beta_0 = \frac{a}{b}, \rho = \frac{r}{b}$.

For the reference configuration, $\beta_0 = 0.5, 0.5 \leq \rho \leq 1, p = 50$ MPa.

The radial, circumferential, and axial stresses were evaluated using the analytical expressions derived in Section 2. These stress components were subsequently substituted into the three-dimensional Tsai–Wu criterion to examine the variation of the failure index across the cylinder wall. The resulting stress and failure-index distributions for the reference configuration are presented in Figure 2.

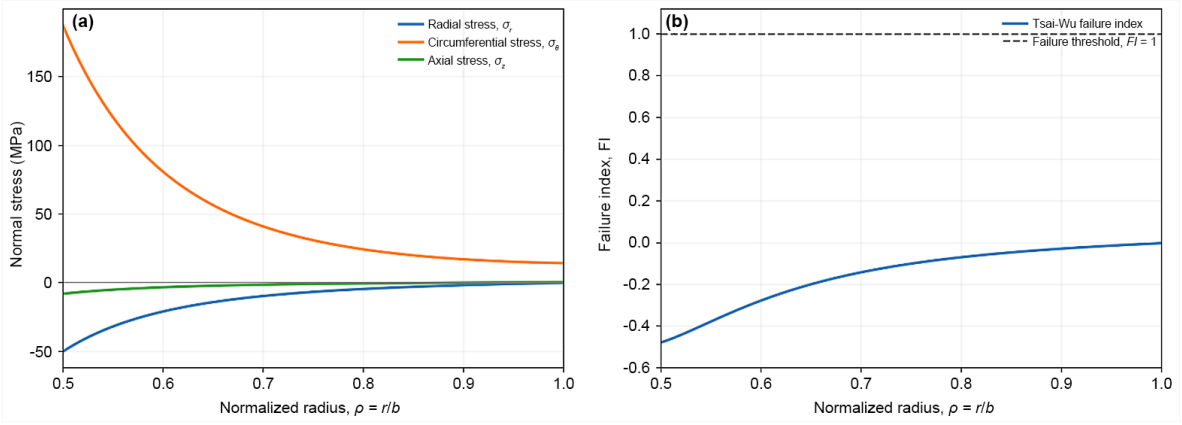


Figure 2. Radial distributions of the normal stress components and Tsai–Wu failure index for the reference cylinder at $\beta_0 = 0.5$ and $p = 50$ MPa

The radial stress satisfied the prescribed boundary conditions, with a compressive value equal to the applied internal pressure at the inner surface and zero radial traction at the outer surface: $\sigma_r(0.5) = -50$ MPa, $\sigma_r(1) = 0$.

The reported calculations showed that the circumferential stress was tensile and decreased substantially across the cylinder wall. For the reference configuration, its maximum value was approximately 188 MPa at the inner surface, decreasing to about 15 MPa near the outer surface. This pronounced variation is associated with the combined influence of cylinder geometry and the directional elastic stiffness of the composite. The circumferential-to-radial modulus ratio was:

$$\frac{E_\theta}{E_r} = \frac{142}{10.3} \approx 13.79$$

The characteristic exponent, calculated from the effective stiffness coefficients, was approximately 3.703. This value reflects the pronounced difference between the radial and circumferential elastic responses of the equivalent orthotropic material.

Under the plane-strain condition, the axial strain was constrained to zero, while a nonzero axial stress developed through the orthotropic constitutive response. For the reference configuration, the axial stress at the inner surface was approximately –7.9 MPa, as obtained from the radial and circumferential stress components using Eq. (3). This contribution was retained in the three-dimensional Tsai–Wu failure assessment.

The calculated Tsai–Wu failure index varied from approximately –0.47849 at the inner surface to –0.00234 at the outer surface. The index increased with radial position but remained below the initial failure threshold throughout the cylinder wall. The negative values resulted from the combined contributions of the linear and quadratic terms in the Tsai–Wu criterion under the specified multiaxial stress state. Although the maximum circumferential stress occurred at the inner surface, the largest failure index at 50 MPa was obtained at the outer surface. This distinction indicates that the location of maximum circumferential stress does not necessarily coincide with the location of the maximum failure index at a given pressure.

This indicates that the reference stress state did not reach the initial failure condition according to the adopted Tsai–Wu criterion. It does not, however, establish an independently validated safety factor or ultimate pressure capacity. The quantitative failure assessment remains dependent on the material strength parameters and interaction coefficients used in the calculation.

The influence of wall thickness was subsequently investigated by varying the inner-to-outer radius ratio while retaining the same material representation and loading framework. Increasing β_0 corresponds to a reduction in wall thickness when the outer radius is held constant. Figure 3 presents the reported variations in circumferential stress and failure index for the selected geometric configurations.

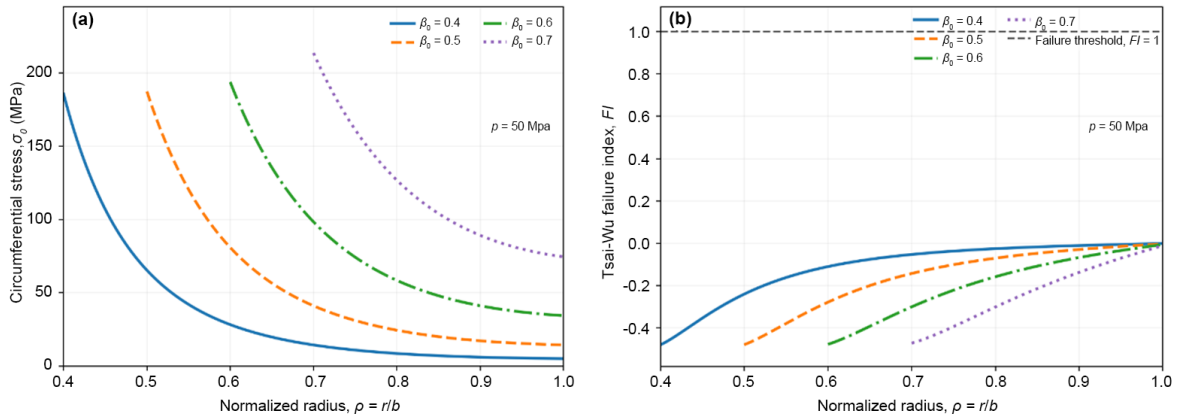


Figure 3. Influence of the inner-to-outer radius ratio on circumferential stress and the Tsai–Wu failure-index distribution under internal pressure

The reported results showed that the maximum circumferential stress increased as the cylinder wall became thinner. For example, the maximum stress near the inner surface was approximately 185 MPa at $\beta_0 = 0.4$, whereas it exceeded 210 MPa at $\beta_0 = 0.7$.

These results indicate that the geometric ratio has a substantial influence on the local stress concentration in the pressure-bearing composite cylinder. A thicker wall generally provides a larger load-bearing section and reduces the circumferential stress under otherwise comparable conditions. The magnitude of this reduction, however, depends on the orthotropic stiffness parameters and the imposed boundary conditions.

The effect of wall thickness on initial failure was assessed through the corresponding Tsai–Wu failure-index distributions. Unlike a criterion based exclusively on circumferential stress, the three-dimensional formulation accounts for the simultaneous contributions of radial, circumferential, and axial stresses. Therefore, the radial location at which the failure index first reaches its critical value must be determined independently for each geometric configuration.

The allowable internal pressure was defined as the smallest positive pressure at which the failure index reached unity at any radial position:

$$p_{\mathrm{cr}} = \min_{\rho \in [\beta_0, 1]} \{ p > 0 : FI(\rho, p) = 1 \}$$

For the linear elastic model, the normal stress components are proportional to the applied internal pressure. At a fixed radial position, the Tsai–Wu failure index can therefore be expressed as a quadratic function of pressure:

$$FI(\rho, p) = A(\rho)p + B(\rho)p^2$$

where, $A(\rho)$ contains the contributions of the linear strength coefficients and $B(\rho)$ contains the quadratic and cross-interaction contributions. Both functions depend on the orthotropic material properties, cylinder geometry, and selected Tsai-Wu coefficients.

The initial failure pressure at a given radial position is obtained by solving:

$$B(\rho)p^2 + A(\rho)p - 1 = 0$$

The smallest positive root satisfying the adopted failure criterion is then compared across the cylinder wall to identify the predicted critical internal pressure. This procedure provides a systematic means of determining the first attainment of the Tsai-Wu failure threshold without assuming in advance that failure must initiate at the location of maximum circumferential stress.

For the reference cylinder configuration, the critical internal pressure predicted by the three-dimensional Tsai–Wu criterion was 170.232 MPa. Initial failure was predicted at the inner surface, where the failure index first reached unity. At an internal pressure of 169 MPa, the maximum failure index was 0.96603, indicating that the predicted initial failure threshold had not yet been reached. The corresponding stress distributions and failure-index variation are presented in Figure 4.

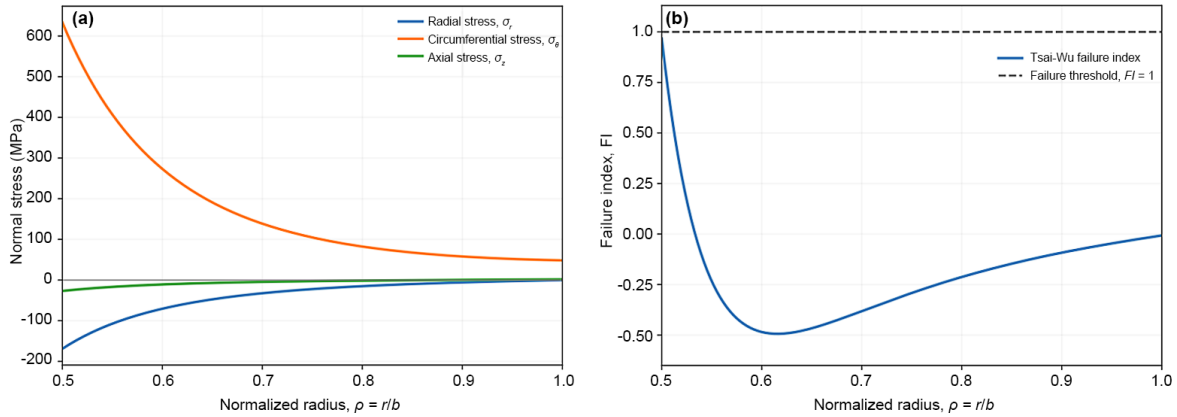


Figure 4. Radial distributions of the normal stresses and Tsai–Wu failure index at an internal pressure of 169 MPa for the reference cylinder configuration

The comparison between the reference loading condition and the higher-pressure case illustrates the increasing importance of multiaxial stress interaction as the applied pressure approaches the predicted initial failure threshold. Since the stress components increase proportionally with pressure in the linear elastic formulation, their combined contributions to the Tsai–Wu index change through both first- and second-order pressure terms. The resulting failure response is therefore not necessarily proportional to the applied pressure.

From an engineering perspective, the analytical formulation provides a means of examining the relationship between wall thickness, orthotropic stiffness, multiaxial stress distribution, and initial failure capacity. Increasing wall thickness may reduce local stress concentrations and increase the predicted pressure required to reach the failure threshold, although this must be weighed against the additional structural mass. The present analysis does not establish an optimized cylinder design, since no formal mass–strength optimization was performed. Instead, it provides a mechanical basis for evaluating geometric alternatives and identifying the conditions under which the selected failure criterion predicts the onset of material failure.

The numerical results should be interpreted within the assumptions of homogeneous orthotropic elasticity, axisymmetric internal loading, and constrained axial deformation. The analysis does not account for individual ply failure, progressive damage, delamination, manufacturing defects, or ultimate burst pressure. These limitations are particularly relevant when extending the present analytical predictions to the design of practical composite pressure vessels.

5 Analytical Verification and Engineering Implications

5.1 Consistency of the Analytical Stress Solution

The analytical solution was examined against the radial equilibrium equation, the prescribed traction boundary conditions, and the limiting case of equal effective radial and circumferential stiffnesses. These checks establish the internal consistency of the elastic stress formulation before its application to the failure analysis.

For the internally pressurized cylinder considered in Section 4, the external pressure is zero. The radial stress must therefore satisfy $\sigma_r(a) = -p, \sigma_r(b) = 0$.

Introducing the dimensionless radial coordinate and geometric ratio, $\rho = \frac{r}{b}, \beta_0 = \frac{a}{b}$, the radial stress distribution obtained from the orthotropic elasticity solution can be expressed as:

$$\sigma_r(\rho) = \frac{p\beta_0^{m+1}}{1-\beta_0^{2m}}\left(\rho^{m-1} - \rho^{-m-1}\right)$$

At the inner surface, substitution of $\rho = \beta_0$ gives:

$$\sigma_r\left(\beta_0\right) = \frac{p\beta_0^{m+1}}{1-\beta_0^{2m}}\left(\beta_0^{m-1} - \beta_0^{-m-1}\right) = \frac{p\left(\beta_0^{2m}-1\right)}{1-\beta_0^{2m}} = -p$$

At the outer surface,

$$\sigma_r(1) = \frac{p\beta_0^{m+1}}{1-\beta_0^{2m}}(1-1) = 0$$

Both boundary conditions are therefore satisfied exactly.

The radial and circumferential stresses must also satisfy the axisymmetric equilibrium equation in the absence of body forces:

$$\frac{d\sigma_r}{dr} + \frac{\sigma_r - \sigma_\theta}{r} = 0$$

In dimensionless coordinates, this becomes:

$$\frac{d\sigma_r}{d\rho} + \frac{\sigma_r - \sigma_\theta}{\rho} = 0$$

The corresponding circumferential stress is:

$$\sigma_\theta(\rho) = \frac{mp\beta_0^{m+1}}{1-\beta_0^{2m}}\left(\rho^{m-1} + \rho^{-m-1}\right)$$

Differentiation of the radial stress expression yields:

$$\frac{d\sigma_r}{d\rho} = \frac{p\beta_0^{m+1}}{1-\beta_0^{2m}}\left[(m-1)\rho^{m-2} + (m+1)\rho^{-m-2}\right]$$

Similarly,

$$\frac{\sigma_r - \sigma_\theta}{\rho} = -\frac{p\beta_0^{m+1}}{1-\beta_0^{2m}}\left[(m-1)\rho^{m-2} + (m+1)\rho^{-m-2}\right]$$

Adding these expressions gives:

$$\frac{d\sigma_r}{d\rho} + \frac{\sigma_r - \sigma_\theta}{\rho} = 0$$

The radial and circumferential stress fields thus satisfy the governing equilibrium equation identically.

A further check is obtained by considering the limiting case in which the effective stiffness coefficients satisfy $C_{11} = C_{22}$.

The characteristic exponent then reduces to:

$$m = \sqrt{\frac{C_{22}}{C_{11}}} = 1$$

Substituting this value into the analytical stress expressions gives:

$$\sigma_r(r) = \frac{pa^2}{b^2-a^2}\left(1 - \frac{b^2}{r^2}\right)$$

And

$$\sigma_\theta(r) = \frac{pa^2}{b^2-a^2}\left(1 + \frac{b^2}{r^2}\right)$$

These are the classical Lamé stress distributions for a thick-walled cylinder subjected to internal pressure with a traction-free outer surface. The agreement establishes the correct limiting behavior of the radial and circumferential stress solution. Recovery of the complete isotropic constitutive response additionally requires the corresponding isotropic relations among the elastic constants and Poisson's ratios.

5.2 Evaluation of the Initial Failure Pressure

The analytical stress solution was combined with the three-dimensional Tsai–Wu criterion to determine the pressure at which the initial failure threshold is reached. Under the assumed linear elastic response, the radial, circumferential, and axial normal stresses vary linearly with the applied pressure. They may therefore be written as

$$\sigma_i(\rho, p) = p s_i(\rho), i \in \{\theta, r, z\}$$

where, $s_i(\rho)$ denotes the stress component normalized by the internal pressure.

Substitution into Eq. (20) gives a quadratic relationship between the failure index and applied pressure:

$$FI(\rho, p) = A(\rho)p + B(\rho)p^2$$

The coefficient of the linear pressure term is:

$$A(\rho) = F_1 s_\theta(\rho) + F_2 s_r(\rho) + F_3 s_z(\rho)$$

The quadratic coefficient is:

$$B(\rho) = F_{11} s_\theta^2(\rho) + F_{22} s_r^2(\rho) + F_{33} s_z^2(\rho) + 2F_{12} s_\theta(\rho) s_r(\rho) + 2F_{23} s_r(\rho) s_z(\rho) + 2F_{13} s_\theta(\rho) s_z(\rho)$$

These coefficients account for the directional strength limits and the interactions among the three normal stresses. The linear term reflects the tensile–compressive asymmetry of the material, whereas the quadratic term includes both individual stress contributions and cross-interaction effects.

At a specified radial position, the initial failure condition is $FI(\rho, p) = 1$.

Consequently, the pressure must satisfy $B(\rho)p^2 + A(\rho)p - 1 = 0$.

When $B(\rho) > 0$, the positive root is:

$$p_{\mathrm{cr}}(\rho) = \frac{2}{A(\rho) + \sqrt{A^2(\rho) + 4B(\rho)}}$$

This expression is equivalent to the positive root of the quadratic equation, provided that the stated condition is satisfied. For other signs of $B(\rho)$, the existence and admissibility of positive roots must be examined directly.

The critical internal pressure for the complete cylinder is determined by the smallest admissible positive failure pressure over the wall thickness:

$$p_{\mathrm{cr}} = \min_{\beta_0 \le \rho \le 1} p_{\mathrm{cr}}(\rho)$$

The corresponding radial position identifies the location at which the selected Tsai–Wu criterion predicts initial failure. This location must be determined from the complete failure-index distribution and need not coincide with the position of maximum circumferential stress.

The calculation also shows why the initial failure pressure cannot be inferred from a single stress component. Even when the circumferential stress dominates the elastic stress field, the radial and axial stresses contribute to the failure index through their individual strength coefficients and interaction terms. The magnitude and sign of these contributions depend on the directional strength limits and the selected Tsai–Wu interaction coefficients.

The pressure obtained from this procedure represents the onset of failure according to the adopted phenomenological criterion. It does not describe the subsequent redistribution of stresses, the development of matrix cracking or fiber damage, or the ultimate burst pressure of the cylinder.

5.3 Engineering Implications and Limitations

The analytical results illustrate the influence of orthotropic elasticity and wall thickness on the mechanical response of internally pressurized composite cylinders. In contrast to an isotropic material, the directional stiffnesses of the carbon/epoxy composite produce a stress distribution governed by the effective radial and circumferential elastic coefficients. The characteristic exponent in the analytical solution determines the variation of the normal stresses through the wall and therefore influences the stress combinations entering the failure criterion.

The geometric ratio $\beta_0 = \frac{a}{b}$ provides a convenient parameter for examining the effect of wall thickness. For a fixed outer radius, increasing β_0 reduces the wall thickness and changes the stress distribution under a prescribed internal pressure. The calculations presented in Section 4 indicated an increase in maximum circumferential stress as the wall became thinner. This trend is relevant to the preliminary sizing of composite pressure-bearing components, where structural mass must be considered alongside the pressure required to initiate material failure.

The three-dimensional failure assessment is particularly important when axial deformation is constrained. Although the axial strain vanishes under plane-strain conditions, the axial stress generally remains nonzero. Its

contribution to the Tsai–Wu criterion depends on the axial tensile and compressive strength limits and the interaction coefficients involving the axial stress. Neglecting these contributions may alter the predicted initial failure pressure, even when the radial and circumferential stress distributions are accurately determined.

The analysis also identifies several limitations that must be considered when applying the results to practical composite pressure vessels. The cylinder was represented as a homogeneous orthotropic continuum, and the individual composite plies were not modeled explicitly. The formulation therefore does not resolve interlaminar stresses associated with discrete ply interfaces, local matrix cracking, fiber breakage, or delamination. Furthermore, the plane-strain approximation represents constrained axial deformation and does not necessarily describe the axial response of a finite-length pressure vessel with deformable end closures.

The predicted failure pressure is also sensitive to the directional strength parameters and the interaction coefficients adopted in the Tsai–Wu criterion. These coefficients require particular care when material data are transferred from unidirectional laminate tests to an equivalent orthotropic cylindrical structure. Without independent experimental characterization or a detailed laminate-level model, the calculated failure threshold should be interpreted as a prediction associated with the assumed material representation rather than a verified design allowable.

The present analytical framework is therefore most appropriate for examining elastic stress distributions, comparing geometric configurations, and assessing initial failure under the specified loading and constitutive assumptions. Independent finite-element or experimental comparisons would be required to establish its predictive accuracy for practical composite pressure vessels. Further extensions could incorporate explicit laminate architecture, alternative failure criteria, temperature-dependent properties, residual stresses, and progressive damage under static or cyclic loading.

6 Conclusions

This study investigated the elastic stress response and failure initiation of an internally pressurized thick-walled cylinder made of Hexcel AS4/3501-6 carbon fiber/epoxy composite. An analytical formulation based on orthotropic elasticity under plane-strain conditions was combined with the three-dimensional Tsai–Wu failure criterion to examine the through-thickness stress distributions, the interaction among the radial, circumferential, and axial normal stresses, and the influence of wall thickness on the predicted failure response. The analytical stress solution was also checked against the prescribed pressure boundary conditions, the radial equilibrium equation, and the classical Lamé solution in the limiting case of equal effective radial and circumferential stiffnesses.

For the reference configuration with an inner-to-outer radius ratio of 0.5 and an internal pressure of 50 MPa, the maximum circumferential stress was 187.332 MPa at the inner surface and decreased to 14.303 MPa at the outer surface. The radial stress varied from -50 MPa to zero, while the axial stress changed from -7.943 MPa to 0.501 MPa across the cylinder wall. These results confirmed that the plane-strain constraint produced a nonzero axial stress and that the three normal stress components should be considered together in evaluating the failure response of the orthotropic cylinder. The calculated Tsai–Wu failure index increased from -0.47849 at the inner surface to -0.00234 at the outer surface, remaining below the initial failure threshold throughout the wall.

Wall thickness had a marked influence on the circumferential stress distribution. At an internal pressure of 50 MPa, increasing the inner-to-outer radius ratio from 0.4 to 0.7 increased the maximum circumferential stress from 185.554 MPa to 213.548 MPa. The corresponding failure-index distributions also varied with the geometric ratio, although the failure threshold was not reached under the investigated reference pressure. The results showed that the location of the maximum circumferential stress did not necessarily coincide with the location of the maximum failure index at a given pressure. Accordingly, the initial failure condition was determined by examining the complete stress-dependent failure index rather than by considering the circumferential stress alone.

For the reference cylinder, the predicted critical internal pressure was 170.232 MPa, with initial failure occurring at the inner surface according to the adopted three-dimensional Tsai–Wu criterion. At an applied pressure of 169 MPa, the maximum failure index was 0.96603, indicating that the predicted failure threshold had not yet been reached. These results demonstrate the importance of evaluating the pressure-dependent interaction among the three normal stresses when estimating the initial failure capacity of a thick-walled orthotropic cylinder. The analytical formulation provides a means of comparing geometric configurations and assessing the influence of wall thickness on the predicted pressure resistance.

The present findings are restricted to a homogeneous orthotropic material representation, linear elastic response, axisymmetric internal loading, and constrained axial deformation. The predicted critical pressure depends on the adopted directional strength limits and Tsai–Wu interaction coefficients and should not be interpreted as an experimentally verified burst pressure. Individual ply failure, interlaminar damage, progressive degradation, and manufacturing-induced residual stresses were not considered. Further investigations should examine the sensitivity of the predicted failure pressure to the assumed strength parameters and interaction coefficients and establish comparisons with independent numerical or experimental results. The inclusion of explicit laminate architecture, temperature and moisture effects, and progressive damage under static or cyclic loading would allow a more detailed

assessment of composite cylinders under practical operating conditions.

Author Contributions

Conceptualization, methodology, software, validation, formal analysis, resources, supervision, data curation, S.P. and A.S.; writing—original draft preparation, writing—review and editing, visualization, validation, formal analysis, resources, software, G.M. and M.R. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) exclusively for English translation, language editing, grammatical corrections, and improving the clarity and academic presentation of the text. ChatGPT was not used to develop mathematical models, derive equations, perform numerical calculations, generate research data, or formulate scientific conclusions. All scientific contributions were independently developed and verified by the authors, who carefully reviewed the final manuscript and take full responsibility for its accuracy, originality, and scientific integrity.

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